

Kumar Aayush

M.Tech – Research – Energy Systems

•CFD & Thermal Engineering • Combustion Modeling • Fuel Blending & Energy Conversion

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Education

Indian Institute of Technology (ISM) Dhanbad

Expected May 2027

M.Tech, Fuel, Minerals and Metallurgical Engineering (FMME) – CGPA: 8.63/10

Dhanbad, Jharkhand

- **Relevant Coursework:** Computational Fluid Dynamics(CFD), Power Plant Engineering, Fuel & Energy Technology, Alternate Energy Systems, Biofuels, Processing of Liquid and Gaseous Fuels

Kalinga Institute of Industrial Technology

May 2016 – Jul 2020

B.Tech, Mechanical Engineering – CGPA: 8.3/10

Bhubaneswar, Odisha

- * **Relevant Coursework:** Engineering Mathematics, Thermodynamics, Heat Transfer, Fluid Mechanics, Strength of Materials, Machine Design

Vidya Vihar Residential School

Jun 2013 – Jun 2015

Higher Secondary Education (XII) – Percentage: 83.5%

Parora, Bihar

Vidya Vihar Residential School

Mar 2012 – May 2013

Secondary Education (X) – CGPA: 9.8/10

Parora, Bihar

Experience

Nuclear Fuel Complex

28 Jun 2026 – 29 Jul 2026

Intern

Hyderabad,Telangana

- * Modeled hot extrusion, piercing, cold-forming, and finishing of high-strength alloy tubes for nuclear applications using **ANSYS simulation**, evaluating billet geometry, reduction ratio, ram force/speed, deformation, and flow stress across 2-3 process configurations.
- * Assessed extrusion die wear against material selection and lubrication regimes, producing recommendations that informed tool-life and manufacturing-efficiency decisions for the facility's production line.
- * Worked directly within industrial automation and process-control systems at a safety-critical nuclear-fuel manufacturing facility, gaining hands-on exposure to regulatory and quality-control protocols uncommon in academic CFD work.

National Hydroelectric Power Corporation

15 May 2019 – 15 Jun 2019

Vocational Trainee

Balutar, Sikkim

- * Mapped hydroelectric power-generation workflows across turbines, generators, transmission systems, and plant-level control mechanisms during on-site training.
- * Assessed safety, routine-maintenance, and environmental-consideration protocols specific to hydroelectric plant operations.
- * Applied fluid-mechanics and energy-conversion coursework to the turbine and generator systems observed on-site.

National Thermal Power Corporation

15 May 2018 – 15 Jun 2018

Vocational Trainee

Bhagalpur, Bihar

- * Analyzed the operational workflow of a coal-fired thermal power plant, including the boiler–turbine–condenser system and feedwater/ash-handling auxiliary systems.
- * Evaluated safety protocols, emergency procedures, and environmental-compliance practices specific to thermal power generation.
- * Connected thermodynamics, fluid-mechanics, and heat-transfer coursework to real-world boiler and turbine performance.

Projects

CFD Simulation of Coal-Biomass Co-Firing and Temperature Distribution in a Boiler | 2026 - Current

- * Building a CFD combustion model in ANSYS Fluent for an existing thermal power plant boiler, with a validated coal-only baseline established as the benchmark for all co-firing scenarios.
- * Simulating 5%-20% coal-biomass blending ratios and comparing their effect on furnace exit gas temperature (FEGT), heat flux distribution, and combustion efficiency against baseline.
- * Evaluating the biomass-substitution threshold at which coal utilization can be reduced while combustion and thermal performance stay within acceptable limits, to support decarbonization planning for coal-fired assets.

Design & Manufacturing of Go-Kart | 2017 - 2020

- * Designed and validated the chassis and drivetrain layout in SolidWorks, integrating a **4-stroke air-cooled engine** with a **wet clutch transmission system**; led fabrication (welding, cutting, machining) through a PVC proof-of-concept to the final frame.
- * Owned material selection and cost estimation across steering, braking, transmission, and frame subsystems, producing a prototype-level feasibility and cost analysis for small-scale manufacturing.

Scientific Reports

Synthesis and Application of CNT Based Energy Storage and Conversion Devices

9 Nov 2021

- * *TechRxiv (Preprint)*
- * Reviewed carbon-nanotube synthesis routes (arc discharge, laser ablation, CVD) and their use as electrode materials in lithium-ion batteries, supercapacitors, and transparent conducting films, comparing performance trade-offs against cost and scalability for mass production.
- * DOI: [10.36227/techrxiv.16926424.v1](https://doi.org/10.36227/techrxiv.16926424.v1)

Heat Transfer Through Brick Walls for Designing the Building Fabric

11 Nov 2021

- * *TechRxiv (Preprint)*
- * Performed transient thermal and CFD analysis of clay brick walls to evaluate phase-change-material (PCM) placement and plywood-based insulation, identifying the optimal PCM position and Chinese plywood as the most effective configuration for reducing heat loss.
- * DOI: [10.36227/techrxiv.16926439.v1](https://doi.org/10.36227/techrxiv.16926439.v1)

Conferences and Seminars

- Attended the Two-Day National Conference on "Net-Zero Emission Technologies for Sustainable Development: Challenges and Opportunities."

February 2026

Technical Skills

- **Simulation & Analysis:** ANSYS Fluent, MATLAB, Life Cycle Assessment (LCA)
- **Design Tools:** SolidWorks (Certified Associate – CSWA)
- **Programming:** Python
- **Core Engineering Areas:** Thermodynamics, CFD, Heat Transfer, Fluid Mechanics, Energy Systems

Achievements

- Certified SolidWorks Associate (CSWA) – Mechanical Design (Dassault Systèmes).
- Completed E-learning course on Self-Charged Hybrid Electric Vehicle – Toyota Kirloskar Motor Pvt. Ltd. & Automotive Skill Development Council (ASDC).
- Represented Purnea District at the State Level, 21st National Children's Science Congress (NCSC).
- German Language Certification – A1 Level (Goethe-Institut).

Social Engagements

Vice-Captain: Kalinga Racers team, involved in the design and manufacturing of a Go-Kart.

Club Member: Automobile Society under KIIT Student Activity Center (KSAC).

Volunteer: National Service Scheme (NSS), Ministry of Youth Affairs and Sports.

Organizer: KIIT Fest 5.0 (Tech-fest) and Khelo India University Games (KIUG).

Sports Engagements: Table Tennis (District level), Volleyball.